

it comes to regulatory issues for obvious reasons.

- **Blockchain is not a one-fit-all model:** Unlike a technology like AI, blockchain is not a one-fit-all model. It caters to a specific set of use cases that mostly revolve around security. So, while the technology might be attractive, it doesn't mean that every tech company has a potential need for it.
- **Problems with integration:** To integrate blockchain into a pre-existing system, a ground up rebuild of the entire system is required. Imagine having a house without any plumbing. Now, to lay pipes, you will have to break all the floors and every wall in the house. It is easier to just break the house and rebuild it from scratch. But what if it wasn't a house but a 1000 acre palace with 100s of rooms? That's basically the conundrum for big tech companies. Even if they found an application for blockchain in their preexisting products, they are just too big to be rebuilt from the ground up.

Regulatory concerns and VC

funding: As already discussed, there is a lack of clarity as to where SEC stands on most crypto projects. Crypto projects are inherently linked to blockchain projects outside of crypto. Every technology requires VC trust in order to thrive as they are the ones who put their money on the line to fund speculative projects. With blockchain, there can be no trust unless there is regulatory clarity. Therefore, the money is running dry, demotivating any further innovation in this space.

The blockchain triad: As stated earlier in the article, the blockchain triad is an existential problem with current blockchains. A technology as pristine as Bitcoin, too, is not immune to this problem. Bitcoin focuses on security which then raises issues with scalability on level 1. Developers bandage this problem by building layer 2 blockchains over the layer 1, but that

adds up complexity and compromises on decentralisation and security. Research is still ongoing as to how this problem can be mitigated, but for now, the problem remains.

The looming quantum

conundrum: As the world edges closer to realising the potential of quantum computing, the very foundation of blockchain technologies is under threat. Quantum computers, with their ability to perform complex calculations at unprecedented speeds, have the potential to crack cryptographic algorithms that keep blockchains secure. Traditional cryptographic methods, like the ones employed in many cryptocurrencies, rely on problems that are computationally hard for classical computers to solve, such as factoring large numbers. However, quantum algorithms, like Shor's algorithm, can solve these problems efficiently. If large scale, practical quantum computers become a reality, they could decrypt blockchain records and forge transactions, effectively dismantling the trust and security that underpin blockchain systems. This quantum conundrum underscores the need for post-quantum cryptographic methods if blockchain technologies are to survive in a post-quantum world.

A potential 'X-chain' that supersedes blockchain altogether: While blockchain has been lauded for its potential to revolutionise sectors from finance to supply chain, it's not impervious to being supplanted. Emerging advancements, such as artificial intelligence (AI), offer enhanced data analysis and decision-making capacities, which might render some blockchain applications

redundant. Quantum computing, as discussed, poses a direct threat to blockchain's cryptographic security. Furthermore, concepts like the 'Blockchain Triad' (which advocates a synthesis of blockchain, AI, and the Internet of Things), hint at a future where blockchain isn't the standalone star but merely a part of a more intricate technological matrix. As history has shown, no technology is sacrosanct. If a more efficient, secure, and scalable solution emerges, it could relegate blockchain to the annals of technological history, much like its predecessors.

Some of the reasons why blockchain is struggling can be addressed rather quickly. For instance, if the SEC moves fast and makes the regulations for blockchain technologies clearer, it can instantly boost the VC trust leading to reflow of funding in the space. This could have a chain effect with big tech trusting the technology and taking steps for adopting blockchain, which could then help startups build disruptive technologies using blockchain. However, there are also some existential threats to blockchain, the solutions for which are currently unclear. Issues such as quantum conundrum and Blockchain Triad are problems that could lead to people developing a newer more advanced technology that renders the concept of blockchain moot.

Personally, I feel blockchain is a great technology with a lot of potential to offer. The rise of Bitcoin is a testament to that. But for blockchain to thrive as a technology at large, the 'big problem' needs to be addressed by the tech community. **END** 🐧

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